

VISION AI PLATFORM · TECHNICAL DOCUMENT

5000-Camera Architecture & Concurrent Incident Handling

This document details how the Code-Ox Vision AI platform is structured for large-scale deployment across 5000 cameras in a multi-region environment, and how it processes multiple simultaneous incidents from a single camera frame — concurrently, atomically, and without data loss.

5000 IP Cameras

4-Tier Multi-Region

~240 Edge Nodes

5 Regions + DR

Parallel by Design

<100ms Alert Latency

ARCHITECTURE · SCALE

The platform uses a 4-tier multi-region federated architecture with disaster recovery. ~240 edge nodes distributed across 5 regions feed into 25 regional aggregators, which connect to a central GPU cluster. Every tier scales horizontally — no architectural changes required as the deployment grows.

CONCURRENCY · INCIDENT HANDLING

Modern detection models return all objects in a single inference pass. Each detected object is evaluated against all configured rules in parallel, and every rule trigger produces its own atomic event with a unique ID, timestamp, and full metadata — with no queueing, no race conditions, and no missed detections.

SCALE TIER
5000 Cameras

ARCHITECTURE
4-Tier Multi-Region

CONCURRENCY MODEL
Parallel by Design

DATE
11 May 2026

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SECTION 01
Platform Overview · Architecture & Concurrent Incident Handling

PLATFORM OVERVIEW

A multi-region platform built for scale and precision.

The Code-Ox Vision AI platform is engineered to operate at city-scale — 5000 cameras across multiple regions — while simultaneously maintaining sub-100ms alert latency and handling any number of concurrent incidents from the same camera frame. Both capabilities are architectural fundamentals, not add-ons.

ARCHITECTURE · MULTI-REGION SCALE

4-Tier Federated Architecture with Disaster Recovery

The platform uses a **4-tier multi-region federated architecture** — the same codebase proven at 1000–2000 cameras, extended with a regional aggregation layer. ~240 edge nodes are distributed across 5 regions, feeding 25 regional aggregators, which connect to a central GPU cluster. A dedicated DR site provides full failover. Every tier scales horizontally without any architectural changes.

CONCURRENCY · PARALLEL INCIDENT PROCESSING

All Incidents in a Frame, Captured Simultaneously

The detection pipeline uses **single-pass inference** — all objects in a frame are returned in one forward pass through the model. Each object is then evaluated against all configured rules in parallel. Every rule trigger produces its own **atomic event** with a unique ID, microsecond timestamp, and full metadata. Multiple incidents in the same frame produce multiple independent events — always, by design.

DESIGN PRINCIPLE

The 5000-camera tier extends the proven 3-tier architecture with a regional coordination layer. Horizontal scaling of node counts and cross-region synchronisation are the only additions — the core detection, rule evaluation, and event dispatch pipeline remains unchanged. Detailed subsystem documentation and BOM available on request.

PLATFORM PERFORMANCE AT SCALE

~12ms

Detection Inference

Single frame · Jetson AGX Orin

≤100ms

Alert Latency

Frame to dashboard

~400k

Daily Events

All cameras · all rules

No Cap

Concurrent Events/Frame

No architectural limit

CONCURRENCY AS A DEFAULT BEHAVIOUR

Multi-object, multi-rule detection is the standard operating mode, not a special case. RF-DETR, D-FINE, and YOLO11/26 return every detected object in a single inference call. The rule engine evaluates all objects against all rules simultaneously. There is no serial bottleneck, no priority queue, and no scenario where one detection blocks another.

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SECTION 02
5000-Camera Architecture · Four Tiers, Five Regions, One Platform

The 5000-camera tier extends the proven 3-tier architecture used at 1000–2000 cameras with a regional layer for multi-region coordination and a dedicated DR site. The architectural shape stays consistent — what changes is horizontal scaling of node counts and the addition of cross-region synchronisation.

TIER 00 · SOURCE

Camera Fleet

5000 IP cameras across 5 regions. RTSP / ONVIF. Any brand, any age. Existing client infrastructure tapped via sub-stream — no rip-and-replace required. Cameras grouped by region for local processing proximity.

TIER 01 · EDGE

Edge Fleet · ~240 Nodes

~240 edge nodes mixed fleet: 140 × Jetson AGX Orin Industrial (18–22 cam each) + 100 × x86 mini-server with RTX 4000 Ada (25–35 cam each). Detection, tracking, and rule evaluation run entirely at the edge — sub-100ms latency, no WAN dependency for alerting.

TIER 02 · REGIONAL

Regional Aggregators · 25 Servers

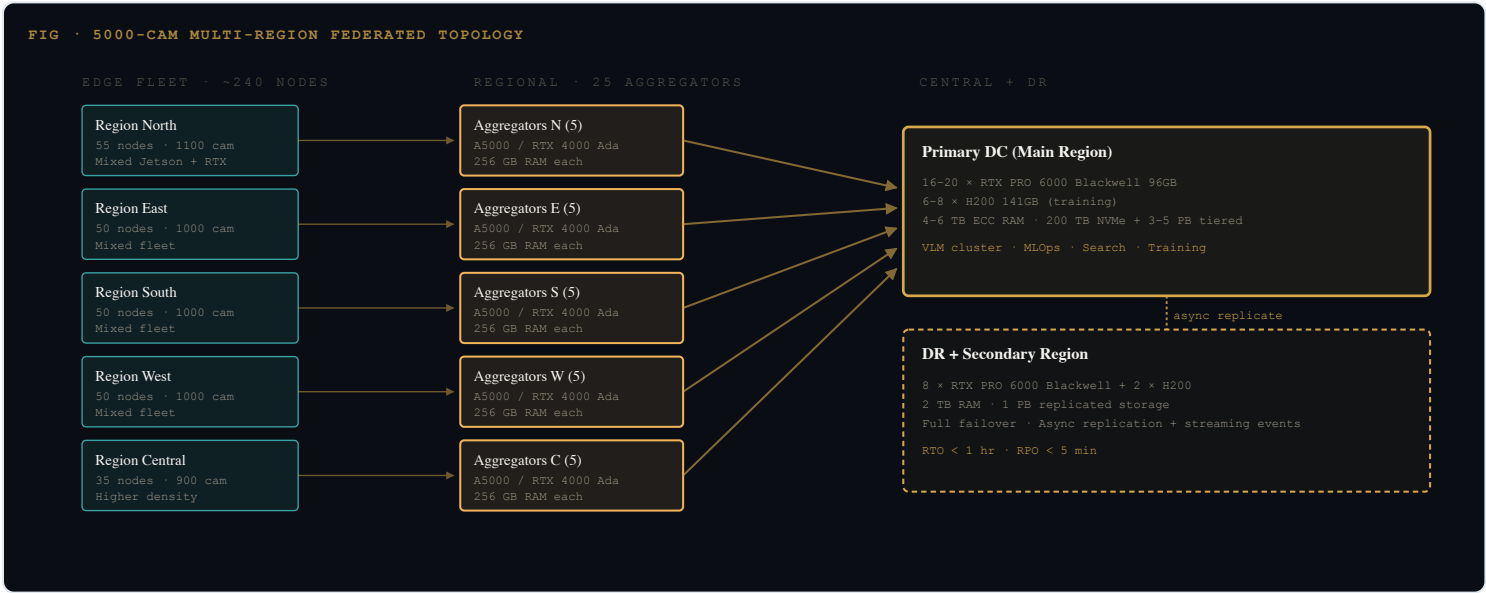
25 aggregator servers (5 per region), each running RTX A5000 / RTX 4000 Ada with 256 GB DDR5 ECC RAM. Responsibilities: cross-camera person re-identification, regional event aggregation, regional event buffer for WAN-outage tolerance, and regional dashboard endpoints.

TIER 03 · CENTRAL + DR

Central GPU Cluster + DR

Primary: 16–20 × RTX PRO 6000 Blackwell 96GB + 6–8 × H200 141GB. 4–6 TB RAM, 3–5 PB tiered storage. VLM inference, semantic search, training, MLOps. DR site with 8 × RTX PRO 6000 + 2 × H200. RTO < 1 hr · RPO < 5 min. Async replication + streaming events.

ARCHITECTURE TOPOLOGY DIAGRAM



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SECTION 03
Infrastructure Breakdown · What's Deployed, End to End

A condensed reference of the physical and software components powering the 5000-camera tier. Detailed BOM with quantities, unit costs, and specifications available separately.

EDGE COMPUTE	~240 edge nodes total, mixed deployment. 140 × Jetson AGX Orin 64GB Industrial SKU for ruggedised sites, 18–22 cam each. 100 × x86 mini-server + RTX 4000 Ada 20GB high-density urban sites, 25–35 cam each.
REGIONAL AGGREGATORS	25 aggregator servers, 5 per region. 2U server + RTX A5000 24GB or RTX 4000 Ada · 256 GB DDR5 ECC RAM each. Cross-camera tracking, regional event aggregation, regional buffer for WAN outage tolerance.
CENTRAL GPU CLUSTER	Primary site: 16–20 × RTX PRO 6000 Blackwell 96GB (VLM inference) + 6–8 × H200 141GB (fine-tuning & training). DR site: 8 × RTX PRO 6000 Blackwell + 2 × H200 with full failover. 4–6 TB DDR5 ECC RAM aggregate. RTO < 1 hr · RPO < 5 min.
STORAGE	3–5 PB tiered storage, multi-region replicated. 200 TB NVMe (hot tier, 24-hr event cache) · 5 PB SAS HDD (30-day continuous, RAID 60) · LTO-9 archive (long-term forensic retention). Geo-distributed object store for cross-region access.
NETWORKING	100/400 GbE backbone, fully redundant. Spine: 4 × 400 GbE switches (Primary + DR). Leaf: 25 × 100 GbE 48-port . Inter-region: MPLS or SD-WAN, 1 Gbps redundant per region.
POWER & COOLING	38–50 kW total IT load, split across regions. UPS: 5 × 100+ kVA N+1 multi-feed (one per region). Cooling: dedicated DC zone, dual CRAC with hot/cold aisle containment, ~30–45% overhead on IT load.
PHYSICAL FOOTPRINT	18–25 racks across multi-row deployment, possibly multi-DC. Primary site: 12–15 racks. DR site: 6–10 racks. Regional aggregator co-locations: 1–2 racks per region.
SOFTWARE PLATFORM	Detection: RF-DETR, D-FINE, YOLO11/26 (TensorRT INT8). VLM: Gemma 4, Qwen3-VL, Llama 4 Scout, InternVL3.5, Mistral Small 4 (on-prem). Tracking: BoT-SORT-ReID + CLIP-ReID for cross-camera re-identification. Data plane: Frigate · Qdrant · ClickHouse · PostgreSQL · MinIO · Kafka. HA stack: Patroni + etcd · ClickHouse Keeper · Keepalived + HAProxy.
PERFORMANCE ENVELOPE	~400K events / day processed across all regions. Peak alert latency < 100 ms (cross-region adds < 30 ms buffer). ~500 Mbps aggregated edge-to-central bandwidth (regionally buffered).
DEPLOYMENT PATH	36–48 weeks total, multi-region phased rollout. Region 1 live first (8–12 weeks), then expansion to subsequent regions in parallel-pair waves. DR site activated mid-rollout.

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SECTION 04

Concurrent Incident Handling · Multiple Incidents, Same Frame, Same Camera

This is one of the most common technical questions about Vision AI platforms — and it's worth answering properly, not casually. The platform is built around the assumption that multi-object multi-rule scenarios are normal, not edge cases.

THE SHORT ANSWER

Yes. Always. Multiple incidents in the same frame produce multiple independent events — concurrently, atomically, with no missed detections and no race conditions.

Modern object detection models — RF-DETR, D-FINE, YOLO11/26 — are **single-pass detectors**. They don't detect "the first object," then "the second object." They take one frame as input and return **all detected objects in a single inference call**, each with its own bounding box, class label, and confidence score.

From that point forward, every detected object is treated independently. Each is evaluated against every configured rule in parallel. Each rule trigger becomes its own event. The architecture has no concept of "one event at a time per camera" — it's parallel by default.

STEP-BY-STEP · WHAT HAPPENS WHEN MULTIPLE INCIDENTS OCCUR IN ONE FRAME

1

Frame arrives at edge node

A single frame from the camera reaches the edge node at the configured rate (typically 5–15 FPS for AI inference, separate from the recording rate). The frame is timestamped with microsecond precision and passed to the detection pipeline.

2

Detection model runs once, returns all objects

The detection model performs a **single forward pass** on the frame. Output is a list of `N` detected objects — each with bounding box, class (person, vehicle, helmet, weapon, etc.), and confidence score. This happens in **8–25 ms** on Jetson AGX Orin depending on model and frame complexity.

3

Tracker assigns persistent IDs across frames

BoT-SORT-ReID + CLIP-ReID associates each new detection with a previously-tracked object if one matches by motion + appearance. If 5 distinct people are in the frame, you get 5 distinct track IDs. Each person is a "tracked entity" carrying state across frames — useful for de-duplication later.

4

Rule engine evaluates all objects against all rules in parallel

For each detected object, the rule engine checks every configured rule simultaneously. **Rules don't run sequentially.** A typical set: *is this person in polygon zone A? Is this person without PPE? Is person count above threshold? Is this person carrying a flagged item?* Each is an independent boolean check, evaluated in microseconds.

5

Each rule trigger becomes an atomic event

Every `(object, rule)` pair that fires becomes its own event with: **unique event ID (UUID v4)**, source camera ID, frame timestamp (microsecond precision), rule triggered, bounding box, track ID, confidence, severity. If 2 people enter restricted zone at once → 2 events. If one person triggers PPE violation AND restricted zone → 2 events (different rules). If 3 people fail PPE in one frame → 3 events.

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Events flow to Kafka bus · downstream consumers process in parallel

Events are written to an internal Kafka event bus. Order is preserved per-camera (FIFO per camera ID), but processing across cameras is fully parallel. Downstream consumers — dashboard, alert router, voice agent, chat dispatcher, ERP webhook — subscribe independently. **One slow consumer never blocks the others.**

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Alert clustering for forensic clarity

If a tracked person triggers the same rule on consecutive frames at 15 FPS, you'd get 15 alerts/second — useless. The clustering layer groups events by `(track_id, rule_id, time_window)` typically 30–60 seconds. **One logical incident card** appears on the dashboard, with all raw events accessible for forensic review. Critical: clustering only applies to **same object x same rule**. Two different people in a restricted zone = two separate clustered incidents — never collapsed into one.

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Dashboard displays concurrent events with separate cards

Each clustered incident appears as its own card in the live event feed, ordered by severity and timestamp. Operators see exactly what happened: which person, which rule, which camera, which moment, with a thumbnail of the triggering frame. Cards can be acknowledged independently.

CONCURRENT MULTI-INCIDENT DISPATCH · VISUAL FLOW



WHY THIS MATTERS FOR SPECTTECH

In a typical deployment, a safety-critical scenario might involve a person entering a restricted zone **without PPE while also triggering a loitering alert** — all in the same frame. The platform captures and dispatches all three rule violations as separate, independently addressable events. Operators can acknowledge each independently, and forensic review can trace the exact frame that triggered each alert.

4.2

SECTION 04.2

Why This Architecture Avoids the Common Failure Modes

Five architectural concerns are commonly raised about concurrent incident handling in Vision AI systems. Each has a deliberate design answer — documented here with the specific mechanism that addresses it.

- ✓

No "first object wins" because detection is single-pass
A common concern: *"if the model sees the helmet violation first, does it miss the weapon?"* The answer is no — modern single-pass detectors return all detections from one inference pass. There is no "first" detection. All objects are returned simultaneously from the same forward pass through the model.
- ✓

No queueing per camera because rule evaluation is parallelised
Rules are evaluated as independent boolean checks across all detected objects. Five rules across three objects = fifteen parallel evaluations, completed in microseconds. There is no serial bottleneck. The rule engine is stateless per evaluation — no shared mutable state, no lock contention.
- ✓

No missed events because the event bus accepts all dispatches atomically
Each rule trigger writes an event to the Kafka bus as an atomic operation. Even if the dashboard or downstream consumer is briefly slow, events accumulate in the bus and are processed in order without loss. The bus provides at-least-once delivery with deduplication at the consumer layer via event UUID.
- ✓

No dashboard overload because clustering happens at the right layer
The clustering layer separates "raw events" (every frame's detections, written to the event store for forensics) from "incident cards" (one logical incident per `track_id` × `rule` × `time-window`). The operator sees actionable cards, not a flood of frame-level detections. The raw event record is always preserved for audit.
- ✓

Cross-camera incidents handled at the Regional layer — not double-counted
If the same person walks across cameras 14 → 15 → 16 in sequence, BoT-SORT-ReID + CLIP-ReID at the Regional Aggregator stitches the track. Cross-camera tracking is a Regional layer responsibility, sitting above per-camera detection — so even concurrent multi-camera incidents are correctly attributed and not double-counted.

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SECTION 05

Performance Envelope · The Numbers, Across the System

Real performance metrics from the architecture at 5000-camera scale. These hold across vertical, deployment, and concurrent-load conditions.

~12ms Detection Inference Single frame · Jetson AGX Orin	<100ms End-to-End Alert Latency Frame to dashboard · safety-critical	~400k Daily Events Processed All cameras · all rules	∞ Concurrent Events / Frame No architectural cap
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LATENCY DECOMPOSITION

Frame capture → edge inference: 8–25ms. Rule evaluation: <1ms. Event bus write: <5ms. Alert routing → dashboard: 10–30ms. Cross-region: adds <30ms buffer. Total safety-critical path: <100ms. Training and VLM workloads are entirely async — they never touch the alerting critical path.

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SUMMARY & NEXT STEPS

Both Questions, Answered with Depth · Happy to Go Deeper on Any Layer

ARCHITECTURE · SUMMARY

4-Tier Multi-Region Federated Architecture with DR

~240 edge nodes across 5 regions. 25 regional aggregators. Central GPU cluster (RTX PRO 6000 Blackwell + H200). Full DR site with RTO < 1 hr and RPO < 5 min. 36–48 week phased regional rollout. The same core platform that operates at 1000 cameras — scaled horizontally, not re-architected.

CONCURRENT HANDLING · SUMMARY

Parallel by Design, No Missed Detections

Single-pass inference returns all detected objects simultaneously. The rule engine evaluates all objects against all rules in parallel. Every rule trigger produces an atomic event. Kafka-backed event bus ensures at-least-once delivery. Alert clustering separates raw events from actionable incident cards without any data loss.

AVAILABLE FOR FURTHER REVIEW

LIVE TECHNICAL DEMO	Walk through any layer in further detail with your technical team. Run live demos against SpectTech's own sample footage — multi-object, multi-rule, multi-camera scenarios.
SUBSYSTEM DOCUMENTATION	Detailed specifications available for: edge node fleet (BOM + specs), regional aggregator configuration, central cluster topology, DR architecture, software stack versions, and HA configurations.
POC / PILOT DESIGN	Can scope a proof-of-concept deployment covering a representative subset of cameras — designed to validate the concurrent incident handling model and architecture with SpectTech's own infrastructure and use cases.
INTEGRATION ARCHITECTURE	API documentation, webhook specifications, and integration architecture for connecting to SpectTech's existing systems — dashboards, ERP, incident management, communication platforms.
DETAILED BOM + PRICING	Full Bill of Materials with quantities, unit costs, and deployment timeline available for the 5000-camera tier. Can be scoped to SpectTech's specific regional topology.

The architecture and concurrent-detection model are designed for city-scale deployments.

Multi-object, multi-rule, multi-camera, multi-region — all handled. We can walk through any layer in further detail with your technical team, run live demos against your sample footage, or share specific subsystem documentation on request.

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